

Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

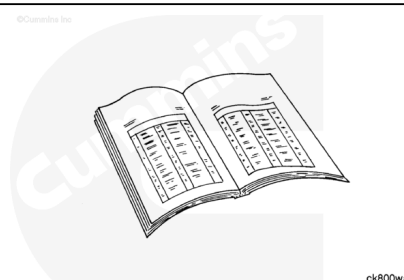
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



ck800wa

- Remove the engine wiring and electrical sensors on the left-bank, as necessary, to gain access to the aftercoolers. Refer to the Troubleshooting and Repair Manual, Electronic Control System, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113.
(/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666113.html)
- Remove the crankcase breathers (if necessary). Refer to Procedure 003-001 (Crankcase Breather (External)) in Section 3. (/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html)
- Drain the cooling system. Refer to Procedure 008-018 (Cooling System) in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the aftercooler assembly. Refer to Procedure 010-002 (Aftercooler Assembly) in Section 10.

(/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)

- Remove the fuel drain line assembly. Refer to Procedure 006-013 (Fuel Drain Lines) in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-013-tr.html)

Remove

QSK45 Engines

One Piece Fuel Manifold.

Note : Steam clean the engine before removing the fuel manifold(s). The exterior of the fuel manifold(s) must be cleaned prior to removal from the engine.

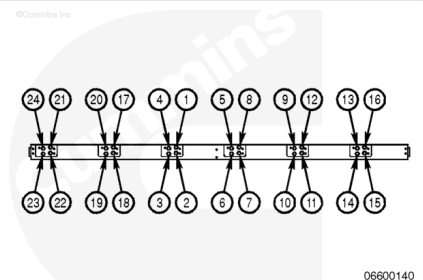
Remove the 24 fuel manifold mounting capscrews.

Remove the four fuel transfer manifold block mounting capscrews.

Remove the left-bank fuel manifold.

Inspect the gaskets for reuse; discard if damaged.

Repeat this process for the removal of the right-bank fuel manifold.



Two Piece Fuel Manifold.

Note : Steam clean the engine before removing the fuel manifold(s). The exterior of the fuel manifold(s) must be cleaned prior to removal from the engine.

Insert two of the lower intake manifold capscrews to support the fuel manifold during removal.

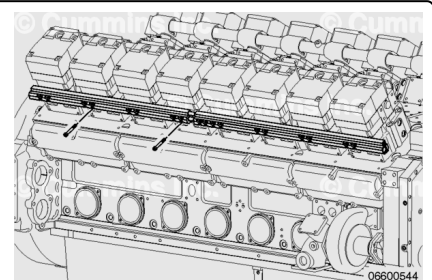
Remove the four fuel transfer manifold block mounting capscrews.

Remove the 24 fuel manifold mounting capscrews.

Remove the left-bank fuel manifold.

Inspect the gaskets for reuse; discard if damaged.

Repeat this process for the removal of the right-bank fuel manifold.



QSK60 Engines

One Piece Fuel Manifold

Note : Steam clean the engine before removing the fuel manifold(s). The exterior of the fuel manifold(s) must be cleaned prior to removal from the engine.

On engines equipped with mechanically actuated injectors, perform the following:

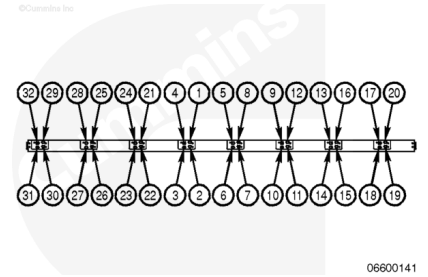
Remove the 32 fuel manifold mounting capscrews.

Remove the four fuel transfer manifold block mounting capscrews.

Remove the left-bank fuel manifold.

Inspect the gaskets for reuse; discard if damaged.

Repeat this process for the removal of the right-bank fuel manifold.



Two Piece Fuel Manifold

Note : Steam clean the engine before removing the fuel manifold(s). The exterior of the fuel manifold(s) must be cleaned prior to removal from the engine.

On engines equipped with mechanically actuated injectors, perform the following:

Insert two of the lower intake manifold capscrews to support the fuel manifold during removal.

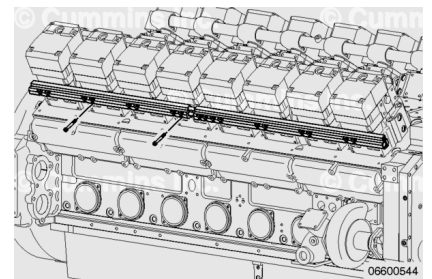
Remove the four fuel transfer manifold block mounting capscrews.

Remove the 32 fuel manifold mounting capscrews.

Remove the left-bank fuel manifold.

Inspect the gaskets for reuse; discard if damaged.

Repeat this process for the removal of the right-bank fuel manifold.



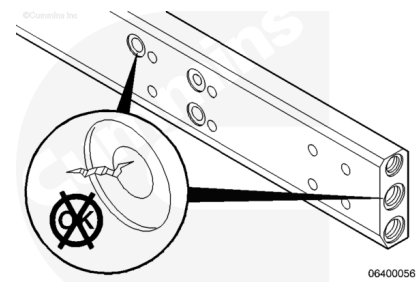
Clean and Inspect for Reuse with Mechanically Actuated Injector

Do **not** remove the straight thread o-ring plugs from the end of the fuel manifold(s), unless evidence of a fuel leak is found.

Inspect the fuel manifold(s) for cracks in the fuel passages.

The fuel manifold(s) **must** be replaced if damaged.

Inspect the fuel manifold gaskets for reuse; discard if damaged.



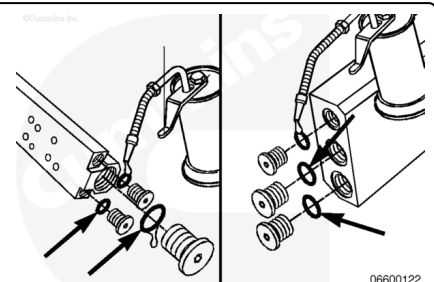
Assemble

with Mechanically Actuated Injector

One Piece Fuel Manifold

If the fuel manifold is being replaced, or the straight thread o-ring plugs from the end of the fuel manifold have been removed for inspection, install the M10, M12, and M27 plugs in the end of the fuel manifold.

Note : Lubricate the o-rings with vegetable oil before installing the threaded plugs into the fuel manifold.



M10 plug	6 n.m	[53 in-lb]
M12 plug	8 n.m	[71 in-lb]
M27 plug	40 n.m	[30 ft-lb]

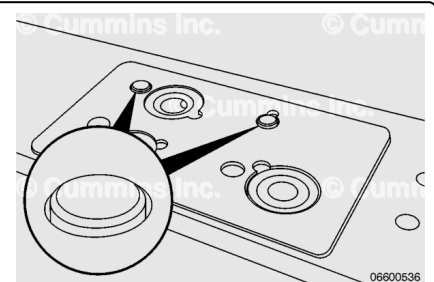
QSK45 Engines

Two Piece Fuel Manifold

Install the fuel manifold gasket onto one of the fuel manifolds in the correct orientation and install the upper two capscrews (as shown) through the rubber embossed holes to be flush with the gasket face.

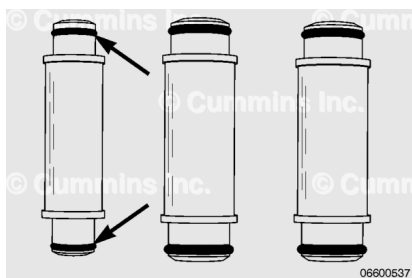
Make sure the capscrews do **not** protrude through the gasket enough to cause interference during installation.

Install the fuel manifold gasket onto the second fuel manifold. Retain the gasket with the capscrews.



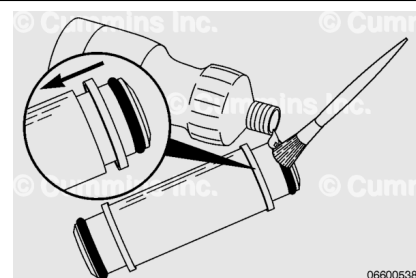
Install new o-rings on the fuel transfer tubes.

Note : New o-rings **must** be installed each time the fuel transfer tubes are removed from the fuel manifold.



⚠ CAUTION ⚠

Push squarely on the fuel transfer tubes when installing them into the fuel manifold. Failure to do so will damage the fuel transfer tube o-rings.



Lubricate the fuel transfer tube o-rings with vegetable oil.

Install the o-rings first, then slide the dust seals up to the fuel transfer tube shoulders.

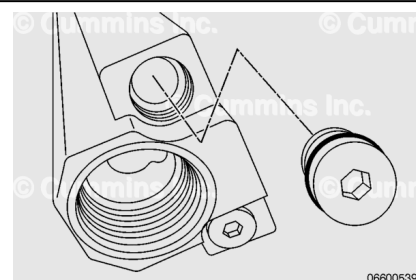
Insert the fuel transfer tubes into one of the fuel manifolds.

Note : The smallest transfer tube is inserted into the bottom hole of the fuel manifold.

If the fuel manifold is being replaced, or the straight thread o-ring plugs from the end of the fuel manifold have been removed for inspection, install the M10, M12, and M27 plugs in the end of the fuel manifold.

Lubricate all threaded plug o-rings with vegetable oil.

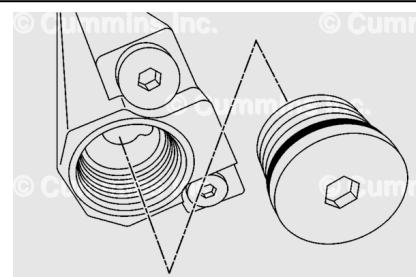
Install one 10mm threaded plug into each end face on the rear of the fuel manifolds and tighten.



M10 plug	6 n.m	[53 in-lb]
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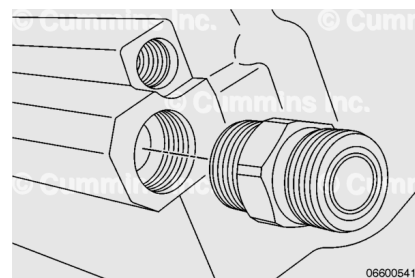
Install one 27mm threaded plug into the rear end face of both fuel manifolds.

M27 plug	40 n.m	[30 ft-lb]
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Install one 27mm male union into the rear end face of both fuel manifolds.

M27 Male Union	40 n.m	[30 ft-lb]
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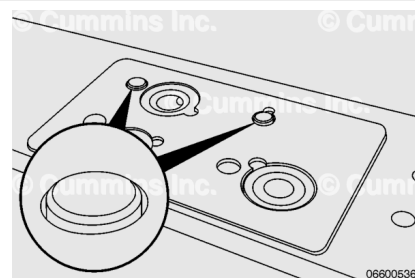
QSK60 Engines

Two Piece Fuel Manifold

Install the fuel manifold gasket onto one of the fuel manifolds in the correct orientation and install the upper two capscrews (as shown) through the rubber embossed holes to be flush with the gasket face.

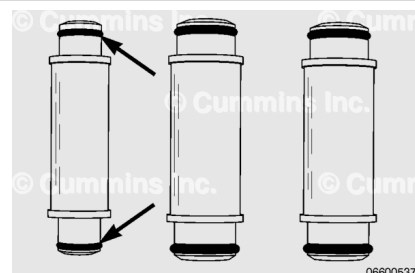
Make sure the capscrews do **not** protrude through the gasket enough to cause interference during installation.

Install the fuel manifold gaskets onto the second fuel manifold. Retain the gasket with the capscrews.



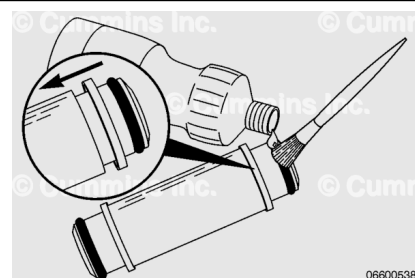
Install new o-rings on the fuel transfer tubes.

Note : New o-rings must be installed each time the fuel transfer tubes are removed from the fuel manifold.



⚠ CAUTION ⚠

Push squarely on the fuel transfer tubes when installing them into the fuel manifold. Failure to do so will damage the fuel transfer tube o-rings.



Lubricate the fuel transfer tube o-rings with vegetable oil.

Install the o-rings first, then slide the dust seals up to the fuel transfer tube shoulders.

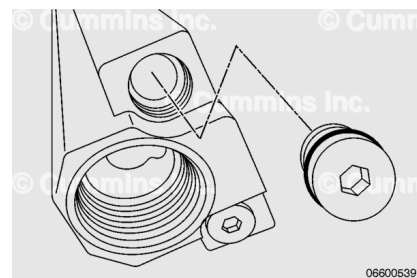
Insert the fuel transfer tubes into one of the fuel manifolds.

Note : The smallest transfer tube is inserted into the bottom hole of the fuel manifold.

If the fuel manifold is being replaced, or the straight thread o-ring plugs from the end of the fuel manifold have been removed for inspection, install the M10, M12, and M27 plugs in the end of the fuel manifold.

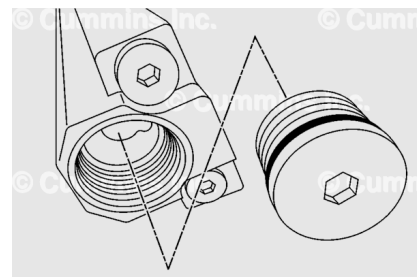
Lubricate all threaded plug o-rings with vegetable oil.

Install one 10mm threaded plug into each end face on the rear of the fuel manifolds and tighten.



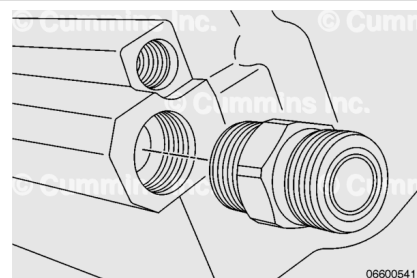
M10 plug	6 n.m	[53 in-lb]
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Install one 27mm threaded plug into the rear end face of both fuel manifolds.



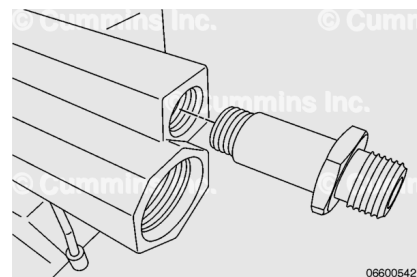
M27 plug	40 n.m	[30 ft-lb]
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Install one 27mm male union into the rear end face of both fuel manifolds.



M27 Male Union	40 n.m	[30 ft-lb]
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Install the timing loop unions.



M10 Male Union	14 n.m	[124 in-lb]
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Install

with Mechanically Actuated Injector

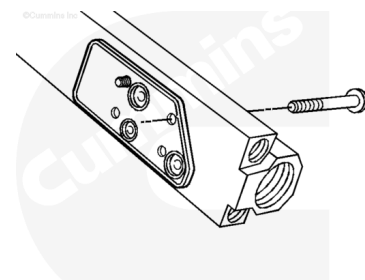
One Piece Fuel Manifold

Use Lubriplate™ Number 105, Part Number 3163086, 3163087 or equivalent, to position and secure the edge-molded gaskets to the back of the fuel manifold.

Install the gaskets to align with each cylinder head.

Insert two capscrews per gasket through the manifold located in the rubber-lined holes of the gasket.

An old style gasket is shown, with old style fuel manifold.



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QSK45 Engines

Once Piece Fuel Manifold

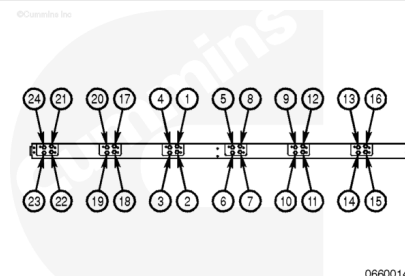
Install the left-bank fuel manifold while making sure the fuel transfer manifold block and tubes are aligned.

Use the sequence illustrated to tighten the 24 capscrews.

Install and tighten the four mounting capscrews to the fuel transfer manifold block. Refer to Procedure 006-010 (Fuel Connection Block (Manifold-Mounted)) in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-010-tr.html)

Torque Value: 9 n·m [80 in-lb]

Repeat the process for the right-bank manifold.



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QSK60 Engines

One Piece Fuel Manifold

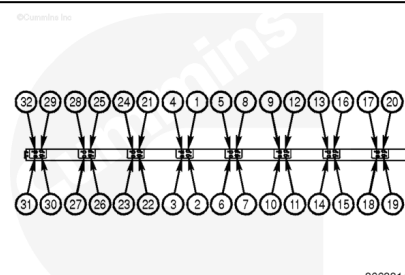
Install the left-bank fuel manifold while making sure the fuel transfer block and tubes are aligned.

Use the sequence illustrated to tighten the 32 capscrews.

Install and tighten the four mounting capscrews to the fuel transfer manifold block. Refer to Procedure 006-010 (Fuel Connection Block (Manifold-Mounted)) in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-010-tr.html)

Torque Value: 9 n·m [80 in-lb]

Repeat the process for the right-bank manifold.



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QSK45 Engines

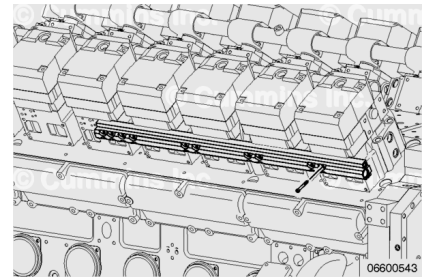
Two Piece Fuel Manifold

Position one of the fuel manifolds up to the cylinder heads and thread in the capscrews.

Insert the remaining capscrews.

Tighten the capscrews.

M6 Capscrew	9.5 n.m	[84 in-lb]
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Insert two of the lower intake manifold capscrews to align the second fuel manifold during installation.

Note : New o-rings must be installed each time the fuel transfer tubes are removed from the fuel manifold.

Lubricate the fuel transfer tube o-rings with vegetable oil.

Install the o-rings first, then slide the dust seals up to the fuel transfer tube shoulders.

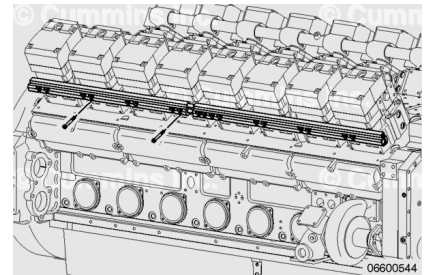
Press the three transfer tubes into the first fuel manifold.

Install the second manifold into the fuel transfer tubes.

Thread in the capscrews and insert the remaining capscrews.

Tighten the capscrews.

M6 Capscrew	9.5 n.m	[84 in-lb]
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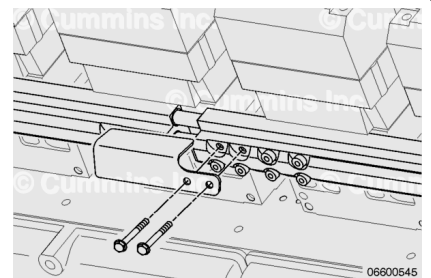


Install the fuel line shield onto the top two capscrew holes of the fuel connection block.

Thread the capscrews in by hand.

Tighten the capscrews.

M6 Capscrew	9.5 n.m	[84 in-lb]
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QSK60 Engines

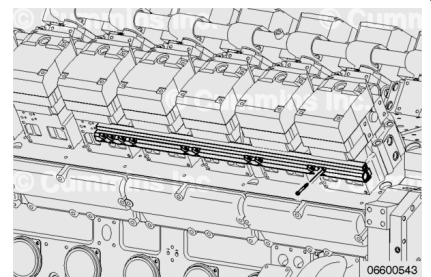
Two Piece Fuel Manifold

Position one of the fuel manifolds up to the cylinder heads and thread in the capscrews.

Insert the remaining capscrews.

Tighten the capscrews.

M6 Capscrew	9.5 n.m	[84 in-lb]
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Insert two of the lower intake manifold capscrews to align the second fuel manifold during installation.

Note : New o-rings must be installed each time the fuel transfer tubes are removed from the fuel manifold.

Lubricate the fuel transfer tube o-rings with vegetable oil.

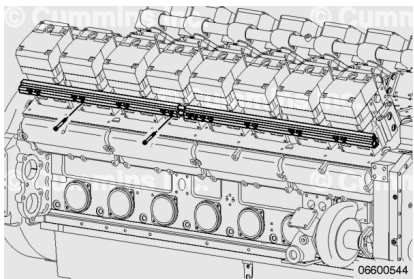
Install the o-rings first, then slide the dust seals up to the fuel transfer tube shoulders.

Press the three transfer tubes into the first fuel manifold.

Install the second manifold into the fuel transfer tubes.

Thread in the capscrews and insert the remaining capscrews.

Tighten the capscrews.

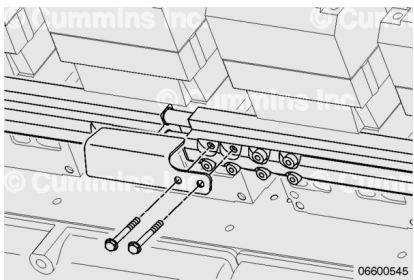


M6 Capscrew	9.5 n.m	[84 in-lb]
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Install the fuel line shield onto the top two capscrew holes of the fuel connection block.

Thread the capscrews in by hand.

Tighten the capscrews.



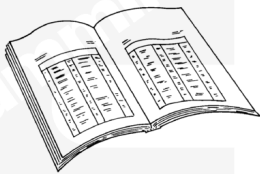
M6 Capscrew	9.5 n.m	[84 in-lb]
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Finishing Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



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- Install the aftercooler assemblies. Refer to Procedure 010-002 (Aftercooler Assembly) in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)
- Install the fuel drain tubes. Refer to Procedure 006-013 (Fuel Drain Lines) in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-013-tr.html)

- Fill the cooling system. Refer to Procedure 008-018 (Cooling System) in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Connect the engine wiring harness and brackets. Refer to the Troubleshooting and Repair Manual, Electronic Control System, QSK19, QSK23, QSK45, QSK60, and QSK78 Engines, Bulletin 3666113.
(/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666113.html)
- Install the crankcase breathers (if necessary). Refer to Procedure 003-001 (Crankcase Breather (External)) in Section 3. (/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html)
- Operate the engine and check for leaks.